How model scores when predictions are made in regions never seen on training

The relation between confidence and accuracy

Experiment

01

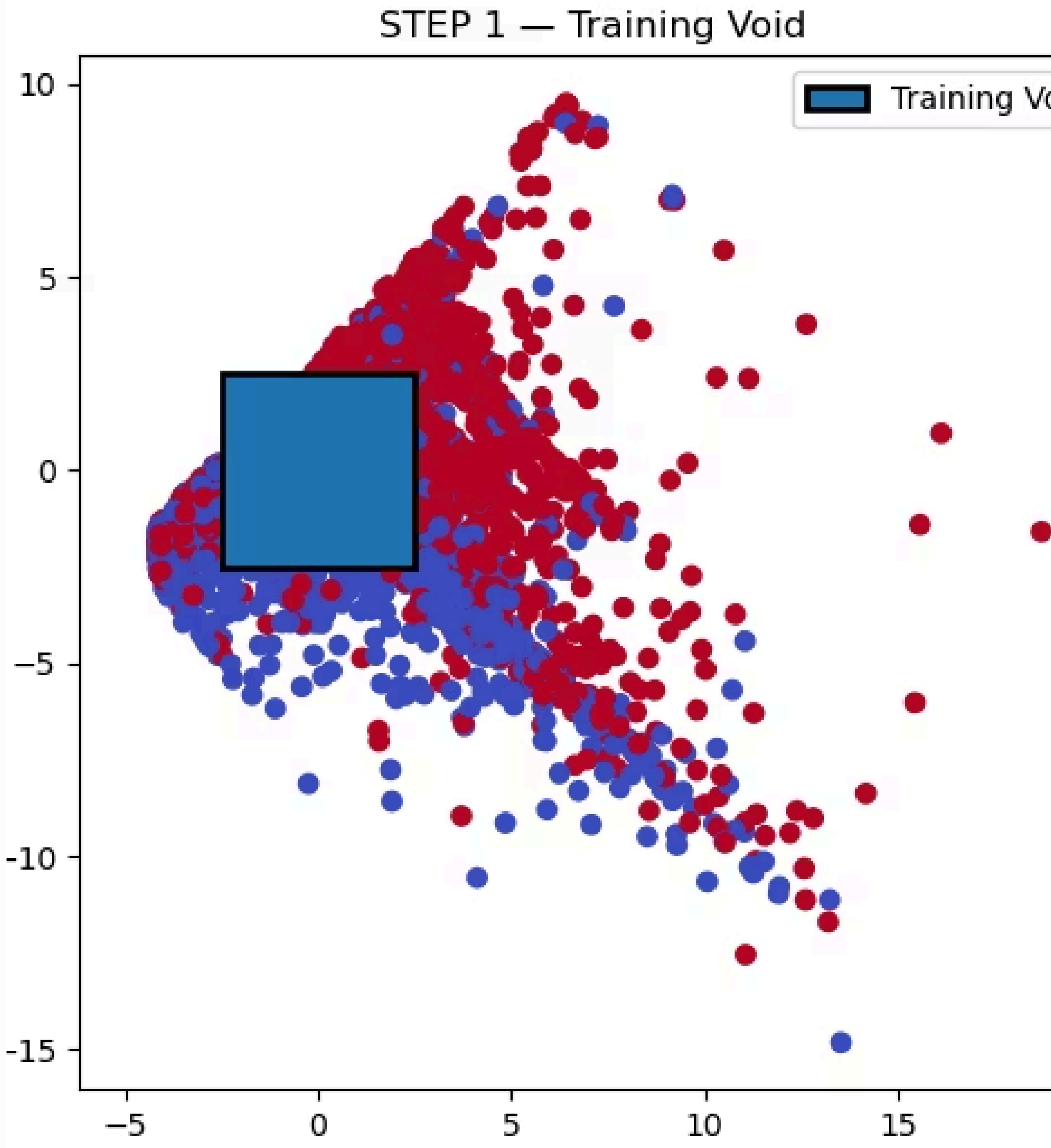
Controlled setup

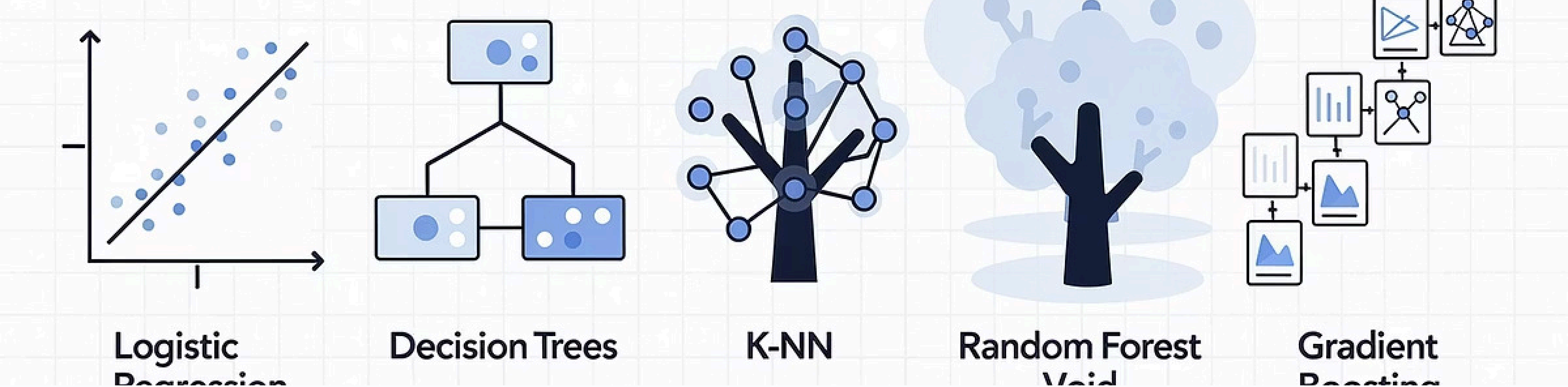
- Used the **UCI Credit Card Default dataset**.
- Reduced data to **2D using PCA** for full geometric visibility.
- Balanced classes to eliminate prior-driven confidence effects.

02

void construction

- Created a **void region inside the feature space**.
- Removed all training samples from this region.
- Test data was left untouched.





Model training

Trained standard classifiers **without special tuning**:

- Logistic Regression
- Decision Tree
- K-NN
- Random Forest
- Gradient Boosting

All models trained on identical data with the same epistemic void.

Evaluation axes

For each test point, computed:

- Correctness
- Model confidence
- Distance to nearest training sample

- **Key analyses**
  - Accuracy vs region
  - Confidence vs region
  - Accuracy vs distance
  - Confidence vs distance

```
Model: Logistic Regression
  ID             | accuracy: 0.668 | mean Conf: 0.673 | N: 1638
  Interpolation  | accuracy: 0.480 | mean Conf: 0.642 | N: 2344

Model: Decision Tree
  ID             | accuracy: 0.653 | mean Conf: 1.000 | N: 1638
  Interpolation  | accuracy: 0.515 | mean Conf: 1.000 | N: 2344

Model: KNN
  ID             | accuracy: 0.707 | mean Conf: 0.782 | N: 1638
  Interpolation  | accuracy: 0.558 | mean Conf: 0.851 | N: 2344

Model: Random Forest
  ID             | accuracy: 0.696 | mean Conf: 0.786 | N: 1638
  Interpolation  | accuracy: 0.551 | mean Conf: 0.741 | N: 2344

Model: Gradient Boosting
  ID             | accuracy: 0.723 | mean Conf: 0.706 | N: 1638
  Interpolation  | accuracy: 0.549 | mean Conf: 0.693 | N: 2344
```

Inference

Confidence does not track correctness

- Accuracy drops sharply in unseen regions.
- Confidence remains largely unchanged.
- This breaks the assumption that confidence reflects reliability.

Distance worsens the failure

As distance from training data increases:

- Accuracy decreases monotonically.
- Confidence stays flat or increases.

The most confident predictions often occur where data density is lowest.

Final takeaway

Predictive confidence is not an uncertainty measure. It is a property of the learned decision function and remains high even in regions where the model has never seen data.